

Riemann–Hurwitz Formula and The Number of Genus

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November 9th, 2025

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Two-Variable Complex Polynomials

We can build a curve $Z(F)$ in $\mathbb{C}^2$ defined by the equation $F(x, y) = 0$. The **genus** is the number of holes in this curve.

Example 1

$$F(x, y) = x^2 + y^2 - 1$$

$Z(F)$ is isomorphic to a sphere with genus 0.

Example 2

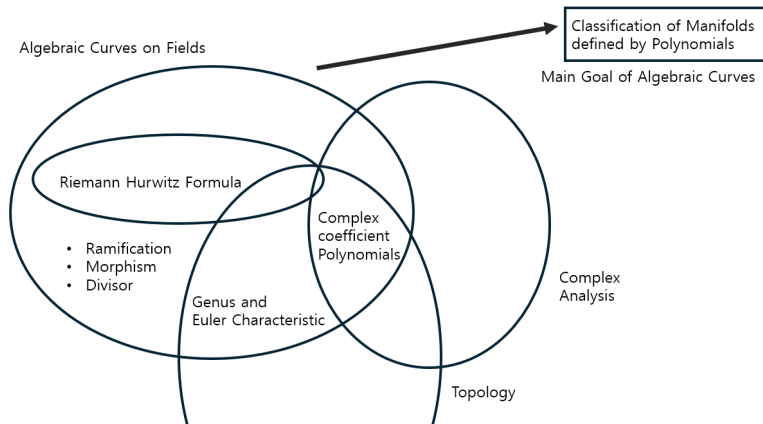
$$F(x, y) = y^2 - x^3 - ax - b$$

$Z(F)$ is an elliptic curve in complex numbers, which is isomorphic to a torus with genus 1.

If there is a two-variable complex polynomial $F(x, y)$, we can build a curve X in $\mathbb{C}^2$ defined by the equation $F(x, y) = 0$. The **genus** is the number of holes in this curve. We can classify polynomials by their genus.

By a morphism, we can transform a curve into another curve. We can calculate the change of genus by the data about polynomial and morphism by **Riemann-Hurwitz Formula**.

Overview



Affine n - space

Let k be fixed field which is algebraically closed. An *affine n -space* over k denoted as $\mathbb{A}^n = \mathbb{A}_k^n$ is defined as follows.

$$\mathbb{A}^n := \{(a_1, a_2, \dots, a_n) : a_i \in k\}$$

An element $P \in \mathbb{A}^n$ will be called a *point*, and if $P = (a_1, a_2, \dots, a_n)$, a_i will be called a coordinates of a point P .

Projective n - space

Let k be fixed field which is algebraically closed. An *projective n -space* over k denoted as $\mathbb{P}^n = \mathbb{P}_k^n$ is defined as follows.

$$\mathbb{P}^n := (\mathbb{A}^{n+1} \setminus \{(0, 0, \dots, 0)\}) / \sim$$

where $\sim$ is an equivalence relation defined as follows.

$$(x_1, x_2, \dots, x_n) \sim (x'_1, x'_2, \dots, x'_n) \quad \text{iff} \quad \exists \lambda (\neq 0) \in k \quad \text{s. t.} \quad x'_i = \lambda x_i$$

Algebraic Set

A set $Y \subseteq \mathbb{A}^n$ is an *algebraic set* if there exists $T \subseteq A = k[x_1, x_2, \dots, x_n]$ such that

$$Y = Z(T) := \{P \in \mathbb{A}^n : f(P) = 0 \text{ for any } f \in T\}$$

A set $Y \subseteq \mathbb{P}^n$ is an *algebraic set* if there exists T which is a set of homogeneous polynomials of S where $S = k[x_0, x_1, \dots, x_n]$, such that

$$Y = Z(T) := \{P \in \mathbb{P}^n : f(P) = 0 \text{ for any } f \in T\}$$

Irreducible

A subset Y of a topological set X is *irreducible* if there does not exist two closed proper subsets Y_1 and Y_2 of Y such that $Y = Y_1 \cup Y_2$. Note that empty set is not considered to be irreducible.

Variety

An *affine algebraic variety* is an irreducible closed subset of $\mathbb{A}^n$, and *quasi affine variety* is an open subset of an affine variety.

A *projective algebraic variety* is an irreducible algebraic subset of $\mathbb{P}^n$, and *quasi affine variety* is an open subset of a projective variety.

Note that a dimension of a variety is its dimension as a topological space, which is Zariski Topology, defined by taking the open subsets to be the complements of algebraic sets.

Morphism

Regular

Let Y be a quasi affine variety or projective affine variety. function $f : Y \rightarrow K$ is *regular at point* $P \in Y$ if there is an open neighborhood $U \subseteq Y$ that $P \in U$ and $f = g/h$ on U where g and h are polynomials. (obviously, h is nowhere zero in U) Additionally, f is *regular on* Y if it is regular at every point.

Morphism

For two varieties X and Y on k , $\varphi : X \rightarrow Y$ is a *morphism* if it is a continuous map that for any open set $V \subseteq Y$, and for every regular function $f : V \rightarrow k$, the function $f \circ \varphi$ is regular.

Morphism

Examples

1. For $n \in \mathbb{Z}_{>0}$, the function $\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ defined as follows is a morphism

$$[x : y] \mapsto [x^n : y^n]$$

2. Define an elliptic curve $E := \{(x, y) \in \mathbb{P}^2 : x^3 + ax + b - y^2 = 0\}$. Then the following function $\psi : E \rightarrow \mathbb{P}^1$ is a morphism.

$$(x, y) \mapsto x$$

Insight

We will mainly focus on a morphism $\varphi : X \rightarrow Y$ where X and Y are locus of polynomial in $\mathbb{A}_{\mathbb{C}}[x, y]$ or $\mathbb{P}_{\mathbb{C}}[x, y]$.

Local Ring and Function Field

Local Ring

Y is a variety and $\mathcal{O}_Y$ is the ring of all regular functions on Y . For a point $P \in Y$, the *local ring of P on Y* denoted by $\mathcal{O}_{P,Y}$ is the ring of all *germs* of all regular functions on Y near P .

Function Field

For a variety Y , define a *function field* $K(Y)$ of Y as an equivalence class of pairs $\langle U, f \rangle$ where U is a nonempty open subset of Y and f is a regular function on U . Here, $\langle U, f \rangle \sim \langle V, g \rangle$ if $f = g$ in $U \cap V$.

Note that the elements of $K(Y)$ are called *rational functions*, and this is different with the concept of rational map.

Genus and Euler Characteristics

Genus

For a given curve S , a *genus* of S is a unique g such that

$$S \cong \underbrace{T^2 \# \dots \# T^2}_{g \text{ times}}$$

where T^2 is a torus.

Euler Characteristics

An *Euler Characteristic* χ of a given finite two dimensional CW complex is defined as

$$\chi = V - E + F$$

where $V, E, F = \# \text{vertex}, \# \text{edge}, \# \text{face}$ in cell decomposition.

Genus and Euler Characteristics

Genus and Euler Characteristics

Euler Characteristic is regarded the most crucial topological invariant when analyzing manifolds in topological sense. Here, the following relation between Euler Characteristic χ and genus g is important.

$$\chi = 2 - 2g$$

Ramification

Function Field on Curves

$K(Y)$ is a function field of the curve Y , which is field consisting of rational functions on Y . Functions in $K(Y)$ are defined as $Y \rightarrow \mathbb{A}^1$.

Valuation

Valuation of t at the point P is a measure of how many times f vanishes. We define valuation of t at the point P as

$$v_P(t) = n$$

$n > 0$ when f has a zero of order n at P

$n = 0$ when f is regular and nonzero at P

$n < 0$ when f has a pole of order $-n$ at P

Ramification

Natural Map On Function Fields

For a morphism $\psi : X \rightarrow Y$, we define $\psi^* : K(Y) \rightarrow K(X)$ as a map

$$t \mapsto \psi^*(t) = t \circ \psi$$

Ramification Index

On a morphism $\psi : X \rightarrow Y$, let $P \in X$ and $Q \in Y$ be two points that $\psi(P) = Q$. Consider a local parameterization $t \in \mathcal{O}_{Q,Y}$ that satisfies $v_Q(t) = 1$. Then we define *ramification index on the point P* as

$$e_P = v_P(\psi^*(t)).$$

Insight

When The curve *folds*, $e_P > 1$.

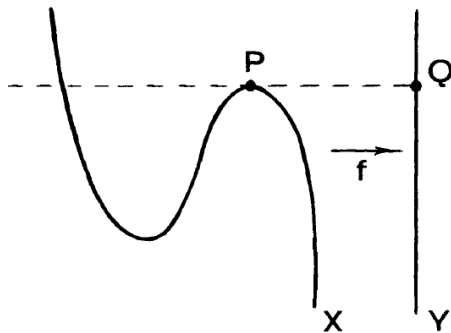


Figure: folded curve maps to a straight curve

When the curve **folds**, $e_P > 1$. Consider a local parameterization $t \in \mathcal{O}_{Q,Y}$. Then while point P' travels on the curve X , the value of $\psi^*(t)$ vanishes to higher order at P , and $\psi^*(t)(P) = 0$. It means that it has $v_P(\psi^*(t)) = e_P > 1$.

Degree of a Morphism

Degree of a Morphism

If $\varphi : X \rightarrow Y$ is a finite morphism of curves, we define *degree* of φ as the degree of the field extension $[K(X) : K(Y)]$

Ramification Index and Degree of a Morphism

For a morphism $\psi : X \rightarrow Y$, for any generic point $Q \in Y$,

$$\deg \psi = \sum_{P \in \psi^{-1}(Q)} e_P$$

Singularity

Let Y be any variety. Y is *nonsingular at a point* $P \in Y$ if the local ring $\mathcal{O}_{P,Y}$ is a regular local ring. P is *singular* if it is not nonsingular.

Divisor

Let X is a noetherian integral seperated scheme which is regular in codimension 1. We define divisor of f denoted by (f) ,

$$(f) = \sum v_Y(f) \cdot Y$$

Where the sum is taken over all prime divisor of X .

Divisor on varieties of $\mathbb{C}$

D is defined as divisor where,

$$D = \sum_{P \in \text{mero}} n_p [P]$$

n_p is the order of pole of P , and $[P]$ can be regarded as basis element of abelian group defined by the points on the curve.

Example of Divisor

If f has order 2, -1 , -1 at $0, 1, \infty$, respectively,

$$(f) = 2[0] - [1] - [\infty]$$

Note that $f = z^2/(z-1)$ has such order and, order of ∞ can be seen by substituting $w = 1/z$

Riemann Hurwitz Formula

Riemann Hurwitz Formula

Let $f : X \rightarrow Y$ be a finite separable morphism on curves.

$$2g_X - 2 = \deg f \cdot (2g_Y - 2) + \sum_{P \in X} (e_P - 1)$$

Recall that $2g - 2 = -\chi$ where χ is an Euler Characteristic.

Riemann Hurwitz Formula

Relation on Euler Characteristics

$$\chi_X = \deg f \cdot \chi_Y - \sum_{P \in X} (e_P - 1)$$

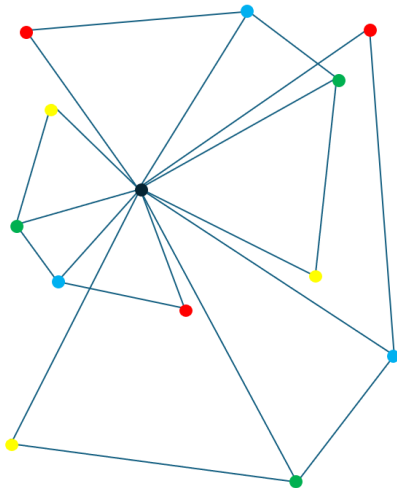
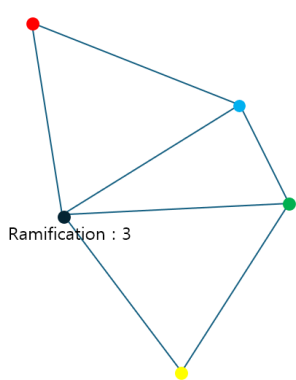
Insight

Since we are focusing on $\mathbb{C}[x, y]$, we can say that X and Y are basically similar with surfaces. So, we can say that $\chi = V - E + F$ for some triangulation on X or Y . Let Π be a triangulation on Y , and Π' be $f^{-1}(\Pi)$.

Ramification index of a point is actually *the multiplicity of a point* in some sense. Therefore, as f is on a algebraically closed field $\mathbb{C}$, we can see that

$|V(\Pi)| = \deg f \cdot |V(\Pi')| - \sum_{P \in X} (e_P - 1)$. Combining with $|E(\Pi')| = \deg f \cdot |E(\Pi)|$ and $|F(\Pi')| = \deg f \cdot |F(\Pi)|$, we can see that the formula is somehow intuitive.

Riemann Hurwitz Formula



Calculating Genus with Riemann Hurwitz Formula

Definition

Hyperelliptic curve is a curve with a form $y^2 = h(x)$ when $h(x)$ is a polynomial.

Using the Riemann-Hurwitz Formula, we are able to calculate the genus of hyperelliptic curves. For a hyperelliptic curve $C : y^2 = h(x)$, think of a map $\psi : C \rightarrow \mathbb{P}^1$, $(x, y) \mapsto x$. Then, check for what P , $e_P > 1$.

When $h(x) \neq 0$, $e_P = 1$ and when $h(x) = 0$, $e_P = 2$.

Each root of $h(x)$ gives a ramification point, and if $\deg(h)$ is odd, there is one additional ramification point at infinity. Since $\deg(\psi) = 2$, genus of $\mathbb{P}^1$ is 0,

$$2g - 2 = 2(2 \cdot 0 - 2) + \sum (e_P - 1) = -4 + r$$

Therefore, when $\deg(h) = 2g + 1$ or $2g + 2$, and genus of C is g .

Rational Curves

Rational Curve

Rational Curve is a curve that can be parameterized by t with rational functions. That is,

$$C : (x(t), y(t)) \text{ for } x(t), y(t) \in \mathbb{C}(t)$$

Genus of Rational Curve

Rational curves has genus 0.

Proof. We consider a map $\psi : C \rightarrow \mathbb{P}^1$, $(x(t), y(t)) \mapsto t$. Then, ψ has different value for every point in C . Therefore, $\deg(\psi) = 1$ and there is no ramification point. By the Riemann–Hurwitz formula,

$$2g - 2 = 1 \cdot (2 \cdot \deg(\mathbb{P}^1) - 2) + 0 = -1$$

Hence, the genus of rational curves is 0.

Classifying Polynomials with Genus

Genus 0

- Rational Curves

Genus 1

- Elliptic Curves

Genus 2+

- Hyperbolic
(nonrational) curve

Classification for $d \leq 7$ in $\mathbb{P}^3$

Degree–Genus Classification of Nonsingular Curves

As the degree d increases, geometric behaviors appear: *from planar curves to complete intersections.*

d	Representative curves in $\mathbb{P}^3$	Genus range g
1	Line $\simeq \mathbb{P}^1$	0
2	Conic (plane)	0
3	Plane cubic; twisted cubic	0, 1
4	Plane quartic; rational / elliptic quartics	0, 1, 3
5	Plane quintic; space curves	0, 1, 2, 6
6	Plane sextic; CI of quadric & cubic (canonical curve)	0–4, 10
7	Plane septic; type (3, 4) on smooth quadric	0–6, 15

- CI = complete intersection. (3, 4) on Q : $d = 7$, $g = 6$.
- Increasing d broadens possible g : a visible hierarchy of complexity.
- Classification captures how algebraic equations manifest as geometric *forms*.

From Classification of Curves to Classification of Varieties

A single polynomial $f(x, y) = 0$ defines an **algebraic curve** C .

Thus, to classify polynomials is, in essence, to classify the **geometric shapes** they generate.

If two polynomials define **birationally equivalent** curves, they share the same **function field** and hence the same **algebraic essence**.

“Classification of polynomials” means classifying **equivalence classes of polynomials** defining the same geometric curve.

From Classification of Curves to Classification of Varieties

Riemann–Hurwitz Formula

$$2g - 2 = \deg f \cdot (2g' - 2) + \sum_P (e_P - 1),$$

The algebraic structure of a polynomial map (degree, ramification) translates into the topological invariant g of the curve.

In higher dimensions, classification relies on richer invariants: **Hodge numbers**, **Picard group**, **Chern classes**, etc. The essence of algebraic geometry: classify all varieties via invariants.

The Three Stages of Classifying Algebraic Varieties

The most ambitious problem: classify all algebraic varieties **up to isomorphism**.

- ① **Classification up to birational equivalence**
- ② **Isomorphism classification of “good” varieties**
- ③ **Measuring the distance from the general case**